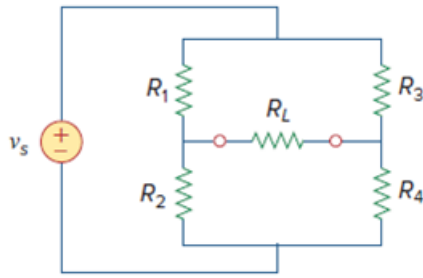


Problem

For the bridge circuit shown in Fig. 4.140, find the load R_L for maximum power transfer and the maximum power absorbed by the load.

Figure. 4.140:



Step-by-step solution

Step 1 of 2

When R_L is removed and V_S is short circuited.

$$R_{Th} = R_1 \parallel R_2 + R_3 \parallel R_4$$

$$R_{Th} = \left[\frac{R_1 R_2}{R_1 + R_2} \right] + \left[\frac{R_3 R_4}{R_3 + R_4} \right]$$

$$R_L = R_{Th} = \frac{(R_1 R_2 R_3 + R_1 R_2 R_4 + R_1 R_3 R_4 + R_2 R_3 R_4)}{[(R_1 + R_2)(R_3 + R_4)]}$$

When R_L is removed and we apply the voltage division principle,

Therefore, $V_{OC} = V_{Th}$

$$V_{Th} = V_{R_2} - V_{R_4}$$

$$V_{Th} = \left[V_s \times \left(\frac{R_2}{R_1 + R_2} \right) \right] - \left[V_s \times \left(\frac{R_4}{R_3 + R_4} \right) \right]$$

$$V_{Th} = V_s \times \left\{ \frac{[(R_2 R_3) - (R_1 R_4)]}{[(R_1 + R_2)(R_3 + R_4)]} \right\}$$

Step 2 of 2

Therefore,

The maximum power is: $P_{Max} = \frac{V_{Th}^2}{4 R_{Th}}$

$$P_{Max} = \left\{ \frac{[(R_2 R_3) - (R_1 R_4)]^2}{[(R_1 + R_2)(R_3 + R_4)]^2} \right\} V_s^2 \left[\frac{(R_1 + R_2)(R_3 + R_4)}{4(a)} \right]$$

Where $a = (R_1 R_2 R_3 + R_1 R_2 R_4 + R_1 R_3 R_4 + R_2 R_3 R_4)$

$$P_{Max} = \frac{[(R_2 R_3) - (R_1 R_4)]^2 V_s^2}{[4(R_1 + R_2)(R_3 + R_4)(R_1 R_2 R_3 + R_1 R_2 R_4 + R_1 R_3 R_4 + R_2 R_3 R_4)]}$$